

SMART UTM — a tensile rig that runs, watches and stops its own tests

Jönköping University · School of Engineering — Materials & Manufacturing

The camera measures what a certified extensometer measures.

20 smart features and 6 closed-loop protocols on a desktop rig. On the same frames, the rig's DIC agrees with a Zwick/Roell XT-205 video extensometer to 0.07 % in scale and 0.19 % in fracture strain — and PLA, PETG and TPU land inside their published bands.

18 / 20

smart features
rig-validated (+2 offline)

0.99928

scale vs XT-205 on the
same frames, R^2 0.9999964

± 1 %

yield and fracture strain
vs the laboratory

12 $\mu\epsilon$

DIC noise floor, 10 s —
reference instrument 45 $\mu\epsilon$

1.78 $\times$ / 1.78 $\times$

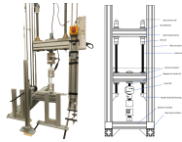
noise vs pixel span
predicted / measured

3

polymers inside their
published bands

THE MACHINE

The rig — built in-house, every control loop closed in software



Two steppers (1.85 Nm, 20:1) drive 5 mm lead screws against a fixed base. An ANYLOAD 3 t cell measures force; a Basler camera measures strain optically at 20.8 px/mm from two markers on the 80 mm gauge. The ESP32 runs STEP/DIR/ENABLE only and knows nothing of test protocols — every protocol, guard and measurement here lives in the PyQt6 app, closing the loop at ~11 Hz.

HOW IT WORKS

One control engine

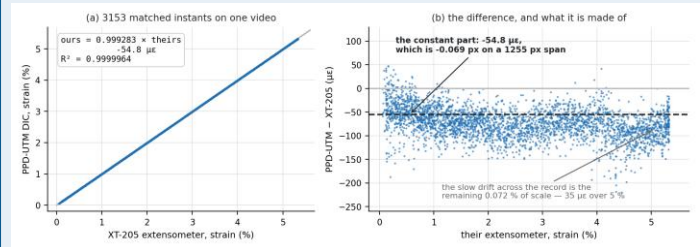
All six protocols are one function called on every load sample (~11 Hz) with load, position and time-matched DIC strain in hand. A protocol only returns a speed — it cannot bypass the shared safety net.

Strain from the specimen

A camera tracks two markers; engineering strain is $(L-L_0)/L_0$ at 20.8 px/mm. The centroid resolves 0.1 px steps at ± 0.02 px.

WHAT THE TESTS SHOWED

Validated against a certified video extensometer — on its own frames



The laboratory's XT-205 video was re-measured by the rig's post-processor: 3153 frames, scale 0.99928, R^2 0.9999964, fracture strain 5.21 vs 5.22 %, and a constant offset of -55 $\mu\epsilon$ — 0.069 px on a 1255 px reference — that vanishes from every slope. On separate specimens the two machines agree within 1 % on yield and fracture strain; the -7.7 % UTS gap is pull rate (2.21 $\times$ the laboratory's), grip seating and batch, not the strain channel.

WHAT IT MEANS

Same frames, same answer → the software is a measurement

Given the XT-205's own frames, the rig's algorithm reproduces the certified record to 0.07 % in scale. Whatever still differs between the two machines is the machine or the specimen, not the strain channel.

The noise is centroid noise → design with pixels

Floor 11.9 $\mu\epsilon$ over 10 s, 26 $\mu\epsilon$ over 900 s, below the reference instrument's 45 $\mu\epsilon$. It scales as $1/P_{\text{Xo}}$: 1675 vs 939 px across the gauge predicted 1.78 $\times$ and measured 1.78 $\times$. More pixels across the gauge is the only lever, and it needs no code.

The crosshead is not the specimen → an extensometer is not optional

Only 19–29 % of early crosshead motion reaches the gauge; a crosshead-based curve overstates strain, and understates modulus, by three to five times.

THE SMART FEATURES, SF1 TO SF20

built & rig-validated (18)

built, verified offline (2)

SF1 Live DIC health HUD — 2 / 2

A green/amber/red badge shows marker lock, tracking % and jitter live — so a dropout is seen during the test, not found later in the CSV

SF4 One-click report — 1 click

A one-page PDF and every plot, recomputed from the CSV by the same library the app runs live

SF7 Stall guard — 0.05 mm / 6 s

Halts a motor that has stopped moving under load

SF10 Auto-preload — 1.03 $\times$

Approaches on a 0.2 → 0.1 → 0.02 mm/s schedule and stops 3 % high, because PLA relaxes about 2 % while held

SF13 Guided wizard — 9 / 9

Connect → camera → data → mode → auto-calibrate → preload → P_{Xo} → prepare → run → save

SF16 Live P_{Xo} overlay — $\Delta = \epsilon$

The frozen reference pair and the live pair are drawn on the feed with a caption whose Δ agrees with the strain readout — the operator sees the measurement, not only a number

SF19 Noise capture — 26 $\mu\epsilon$

Records the instrument at rest and separates offset and drift (correctable) from the residual sd — the uncertainty, which does not subtract

SF2 Prepare specimen — 4 → 1

One button tares position, force and DIC and clears the plots

SF5 Auto-stop at fracture — 1.2 s

A load-collapse detector halts the motor the moment the specimen breaks

SF8 Release load — ≤ 5 N

Taring at preload makes the readout say 0 N while 300 N is still on the specimen

SF11 Auto-metadata link — 3 links

Every save links the CSV, the captured video and the registry row by name and proposes the per-specimen folder — nothing a run produces can be orphaned

SF14 Test registry — 36 runs

Each run indexed with whatever its loading path supports — UTS, ϵ_f , E , σ_y , toughness — plus the force anchor that turns a tared reading into a true load

SF17 Video + image capture — ≈ 47 μs

PNG stills plus three AVI views (raw, contrast-boosted, adaptive speckle) recorded with the run, at a measured cost that never touches the detection budget

SF20 Manual blob selection — 9 / 9

The operator clicks the two markers on a frozen frame and tracking follows the picks with the shape gates off — keeps measuring where automatic detection is deliberately killed

SF3 Recipes / settings — 1 click

Saves and restores an entire setup, including all six protocols' parameters — so repeats are identical by construction, not by memory

SF6 Constant-strain-rate testing — 0.05 %/s

Closed loop on the specimen's own $d\epsilon/dt$: the crosshead adapted itself 0.10 → 0.05 mm/s, fast in the elastic region, slow through necking

SF9 Six closed-loop protocols — 6 / 6

cyclic · staircase · relaxation · creep · staircase → FRACTURE · progressive-cyclic → FRACTURE

SF12 DIC auto-calibrate — 1 click

Exposure and threshold are swept on the live camera and scored on marker count, contrast and centroid jitter; the winner is applied with the score shown

SF15 Dead-DIC guard + hard backstops — 3.0 $\times$

A controller ramping blind on a stale camera reading spikes the force

SF18 DIC post-processing — 0.99928

Measures strain from ANY recorded video with the rig's own pixel-to-strain rule